Efficient Visualization of Knowledge Bases via Space-Time Graphs: The OHARA Architecture

Thanh-Tung Pham
University of Science, Vietnam National University – Ho Chi Minh City
pttung@fit.hcmus.edu.vn

***Abstract*—Flat-chunk retrieval-augmented generation (RAG) processes content without providing a coherent spatial or temporal dimension for human cognition. Effective knowledge bases must reduce cognitive complexity and imitate the way human beings understand structural relationships. We propose OHARA, a framework structured around a *Space-Time Graph* that unifies document structural hierarchy, temporal decay, and cross-document entity pivots into a single substrate. The offline framework maps document components into a 3D sunburst-tunnel visualization (time $\rightarrow$ Z-axis, ontology $\rightarrow$ sunburst cross-section, document structure $\rightarrow$ radial discs), while the online workflow executes an eight-phase hybrid retrieval sequence fusing lexical, dense vector, ontology, entity, structural, and cross-document candidate signals. Evaluated across QASPER (200 academic papers) and MultiHop-RAG (609 news articles) corpora, our tuned pipeline reaches ranking parity with single dense retrieval on QASPER and a marginal MRR gain on MultiHop-RAG (0.812 vs. 0.811), while establishing a corroboration-gated Principal tier. This tier abstains on 45.6% of unanswerable queries (57/125) compared to 0.0% for standard top-k filtering, holding a 91.5% Principal-hit rate on answerable queries. This is an abstention-behavior gain, not a ranking-quality one: removing the constraint moves Hits@10/MRR by under 0.5 points. We additionally run LightRAG and GraphRAG, both reconfigured to OHARA's exact models, as head-to-head baselines on both corpora. The offline ingest pipeline builds the full semantic graph at roughly $12 per 1,000 documents (single-corpus estimate), and the visual rendering scales sub-linearly to 12,323 nodes across five measured points. We report these empirical findings honestly, including small-sample caveats and a tuned-on-test risk in the QASPER weight search, and suggest that OHARA establishes a foundation for seeable and auditable knowledge management.**

***Index Terms*—retrieval-augmented generation, knowledge graph, information visualization, ontology, temporal reasoning**

I. Introduction

Flat-chunk RAG pipelines segment documents into linear chunk sequences, stripping away macro-level structural context and temporal dimensions. This design forces large language models and human users to navigate context walls, compounding memory degradation effects [1]. Over years, retrieval-augmented generation has expanded into diverse graph-structured frameworks [5], with graph-based variants surveyed as an emerging subfield [6]. However, current architectures often over-complicate graph construction without addressing how human cognition processes spatial and temporal hierarchies. OHARA targets structured domains such as research papers, legal filings, and technical manuals by providing a unified Space-Time Graph.

An ideal retrieval framework should imitate the way human beings understand organized corpora: mapping high-level concepts down to granular text segments. We explicitly structure our contribution to balance retrieval efficiency with transparent inspection.

Contributions.

- **3D Sunburst-Tunnel Visualization Component:** A visualization framework mapping temporal evolution to the Z-axis, ontology classifications to a polar sunburst plane, internal document structure to radial discs, and decay class to temporal reach.
- **Space-Time Graph Substrate:** A data model unifying structural hierarchy, temporal decay, and cross-document entity pivots to represent context at the level of human knowledge concepts.
- **Tiered Hybrid Retrieval Engine:** An eight-phase retrieval sequence designed to extract, filter, normalize, and evaluate candidate signals, achieving ranking parity with dense vector retrieval while providing explicit provenance.
- **Corroboration-Gated Abstention:** A gating mechanism operating on a threshold of $\geq$2 independent signal sources. The Principal tier abstains on 45.6% of unanswerable queries (57/125) on MultiHop-RAG versus 0.0% for plain top-k cutoffs, retaining 91.5% accuracy on answerable candidates; this is an abstention-behavior effect, distinct from ranking quality, which the constraint changes by under 0.5 points.
- **Empirical Baseline Comparison:** We run LightRAG and GraphRAG, both reconfigured to OHARA's exact Gemini models, head-to-head on both QASPER and MultiHop-RAG, addressing the absence of direct empirical comparison against cited graph-RAG systems.
- **SUMO-Grounded Semantic Layer:** An ontology mapping pipeline that anchors visualization coordinates and optimizes tag-expansion candidate retrieval.

Systems focused solely on ranking metrics miss this: knowledge bases also need auditability and human-centric spatial navigation. OHARA does not claim graph-augmented retrieval outranks dense vector baselines; our evaluation (Sec. VII)

TABLE I
CAPABILITY COMPARISON AGAINST GRAPH-RAG BASELINES.

Capability	GraphRAG	LightRAG	HippoRAG	OHARA
Entity/relation subgraph	✓	✓	✓	✓
Structural hierarchy (DoCO)	–	–	–	✓
Cross-doc similarity edges	partial	✓	✓	✓
SUMO ontology grounding	–	–	–	✓
Temporal decay scoring	–	–	–	✓
3D space-time visualization	–	–	–	✓
Tiered explainability	–	–	–	✓

shows the hybrid pipeline matches the best single signal while rendering a navigable coordinate map.

II. RELATED WORK

Graph-based RAG architectures (GraphRAG [7], LightRAG [8], HippoRAG [9]) extract entity-relation subgraphs to link cross-document concepts. Beside of those, explainable retrieval systems (Self-RAG [10], Corrective RAG [11]) filter candidates to provide output provenance. However, these frameworks discard explicit document structural hierarchies and lack integrated temporal coordinate systems.

Over years, 2D radial space-filling techniques [2] have successfully visualized complex structural hierarchies. Existing 3D temporal graph visualizers, by contrast, fail to integrate ontology cross-sections with temporal tunnel axes. Table I shows the capability boundaries: OHARA bridges structural hierarchy, ontology grounding, temporal scoring, and spatial visual mapping within a single optimized workflow.

III. THE SPACE-TIME GRAPH DATA MODEL

We model the knowledge base as a 5-tuple Space-Time Graph: $G = (V, E, \tau, \delta, \sigma)$. The vertex set $V = V_{docs} \cup V_{sect} \cup V_{para} \cup V_{tab} \cup V_{ent}$ partitions candidates into document roots, structural section nodes, content paragraphs/tables, and named entities. Directed edges $E \subseteq V \times V \times R$ are defined by seven relation types

$$R = \{\text{HAS\_CHILD}, \text{NEXT\_SIBLING}, \text{BELONGS\_TO}, \text{MENTIONS}, \text{RELATED\_TO}, \text{SIMILAR\_TO}, \text{TOC\_REF}\},$$

spanning structural tree, semantic cross-cutting, and Jaccard-gated cross-document relationships.

The offline stage assigns temporal mapping $\tau : V_{docs} \rightarrow \Theta$ along the visual Z-axis and classifies decay parameters $\delta : V_{docs} \rightarrow D$ across four classes (EVERGREEN $\lambda{=}10^{-6}$, SCHOLARLY $\lambda{=}10^{-4}$, CURRENT $\lambda{=}10^{-2}$, EPHEMERAL $\lambda{=}10^{-1}$). Nodes are promoted to EVERGREEN when the in-degree of SIMILAR_TO edges ≥ 5. Narrative edge enrichment σ attaches action verbs, tag sets, and temporal relations to cross-document links. Temporal decay scoring uses $w \cdot e^{-\lambda \Delta t}$ under a five-layer protection mechanism (no temporal intent, Principal-tier immunity, high-BM25 immunity, weak-candidate-only, and date-range coverage boost).

To reduce computational overhead during query execution, candidate selection follows a coarse-to-fine pre-filtering workflow:

```
BEGIN Document Pre-filtering Workflow
1. FOR ALL d ∈ V_docs DO
2.   Extract paragraph-level
entity_slugs and sumo_tags
3.   Union attributes onto root node d
(Document Rollup)
4. END FOR
5. IF Query arrives THEN
6.   Scan O(n) document rollups,
n = |V_docs|
7.   Prune low-scoring documents before
descending into content chunks
8. END IF
END Workflow
```

This pre-filtering stage reduces candidate evaluation complexity by ~50–60× at standard corpus densities (16 sections, 44 paragraphs per document), allowing rapid computation prior to fine-grained structural descent.

IV. MULTI-PHASE HYBRID RETRIEVAL ENGINE

The online retrieval engine executes an eight-phase candidate processing sequence:

- **Phase 0 (Fingerprinting):** Classify query mode, extract SUMO tags, entity hints, and temporal constraints for phrase-length inputs.
- **Phase 0b (TOC Selection):** Execute LLM traversal over seed document tables of contents to establish structural entry points.
- **Phase 1 (BM25 Full-Text):** Extract lexical match scores across content nodes.
- **Phase 1b (SUMO Overlap):** Filter candidates via ontology hierarchy tag expansion.
- **Phase 1c (Multi-Hop Traversal):** Traverse SIMILAR_TO edges carrying narrative verbs and summaries.
- **Phase 1d (Dense Vector Similarity):** Evaluate candidate cosine similarity via `gemini-embedding-2` (768-d).
- **Phase 2 (Entity Pivot):** Segment and pivot across shared canonical entity nodes.
- **Phase 3 (Structural Traversal):** Traverse local structural tree (depth 2) with corrective zero-SUMO-overlap filtering.

Phase 4 fuses max-normalized phase scores via weighted summation (base weights: BM25 1.0, vector 0.5, SUMO 0.4, entity 0.6, cross-doc 0.4, structural 0.3) scaled by query mode multipliers, integrating guarded temporal contributions.

Candidates are categorized into explicit functional tiers to imitate human verification patterns:

- **Principal Tier:** Candidates requiring $\geq$2 independent signal contributions (typically `bm25∩vector`), score above the 75th percentile threshold, and carry cross-document evidence.
- **Integrity Tier:** Direct structural or cross-document neighbors attached to Principal nodes.
- **Explorer Tier:** Metadata-only frontier nodes evaluated within a weakening-scent weight band.

Unlike standard top-k cutoffs, multi-signal corroboration lets the system abstain explicitly when evidence is insufficient, which limits hallucination.

V. Ingest Pipeline

The ingestion workflow is split into two distinct execution halves:

First Part: Offline Parsing & Structuring. Input formats (PDF, EPUB, DOCX) are parsed via LiteParse and converted into DoCO node hierarchies using content-hash-cached LLM processing (`gemini-2.5-flash-lite`, concurrency 4).

Second Part: Semantic Purification & Deduplication. Extracted SUMO tags undergo a three-stage validation against a 22,700-entry index (exact, case-insensitive, alias lookup). Entities are normalized and deduplicated across documents. Jaccard similarity thresholds trigger cross-document edge creation, enriched with LLM-generated narrative summaries.

At scale, ingesting 200 QASPER papers consumed 6.2M prompt and 4.5M output tokens (~53k tokens/doc), totaling \$2.41 (~\$12 per 1,000 documents). This is a single-run estimate from one corpus; the MultiHop-RAG run (Sec. VII) gives a second point at a different per-document token rate, but two runs on two corpora do not bound variance across document types or sizes. Content-hash caching guarantees idempotent re-ingest, ensuring rapid computation and near-zero cost upon retry. Forced re-ingest of partial documents duplicated structural edges (23k duplicates across 13k pairs), indicating idempotency holds at chunk level rather than edge level.

For reference, indexing the same corpora with GraphRAG [7] (same Gemini models, community-report generation included) cost \$21.35 per 1,000 documents on QASPER (\$4.27/200 docs) and \$16.18 per 1,000 on MultiHop-RAG (\$9.84/608 docs, one retry after a transient embedding-quota error), 35–80% more than OHARA's ingest on the same two corpora. This is not evidence that OHARA's pipeline is inherently cheaper than graph-RAG approaches in general: GraphRAG's community summarization step is optional overhead for our local-search-only evaluation and would not be incurred by systems that skip it (e.g., LightRAG, whose ingest cost we did not instrument in this study).

VI. Space-Time Graph Visualization

To map macro corpus organization to micro document details, the interactive 3D visual renderer separates spatial coordinates into three explicit dimensions:

- **Z-Axis (Temporal Dimension):** Positions documents chronologically in resolution-bucketed visual layers.
- **XY Polar Plane (Ontology Dimension):** Maps document root coordinates according to SUMO category classification (θ derived from recursive angular partition, $r = R_{max}\sqrt{depth/D_{max}}$). Topics stack into stable visual columns along Z.
- **Radial Discs (Structural Hierarchy Dimension):** Unfolds each document locally perpendicular to Z. Document nodes occupy the center, section nodes form concentric inner rings, and paragraphs/tables populate the outer perimeter.

Fig. 1. The interactive application. *Top*: full tunnel view at year resolution, 50 documents, colored by document. Sidebar lists selected documents and the node-type legend; header shows live corpus stats and view-mode toggles. *Bottom*: zoomed into a single document's disc with a paragraph node selected, opening the right-hand inspector panel with the section outline and full extracted text/metadata (published date, source, category).

Entities reside at the centroid of mentioning documents, pushed outward by a fixed offset. Semi-transparent decay auras (5/2.5/1/0.4 disc-spans for EVERGREEN/SCHOLARLY/CURRENT/EPHEMERAL) display temporal reach spatially.

Rendering performance relies on Three.js `InstancedMesh` batching (one draw call per shape bucket). Benchmarked on headless WebGL rendering from 713 to 12,323 nodes, scene rebuild latency grew from 1.5 s to 2.3 s while JS heap usage expanded from 52 MB to 80 MB (a 17× node increase causing ~1.7×/~1.5× memory and compute growth). This confirms computational scalability is bounded by visual density rather than geometry generation, though the claim rests on five single-run samples rather than repeated trials, so no variance is reported. Fig. 1 presents the operational application interface, and Fig. 2 illustrates multi-angle visual tunnel projections.

VII. Evaluation

We evaluate OHARA across two corpora: QASPER (200 academic papers, 150 answerable questions, scored corpus-wide over 229 documents including distractors) and MultiHop-RAG (609 news articles, 500 stratified queries including 125 null/unanswerable queries).

TABLE II
QASPER RETRIEVAL QUALITY (150 QUESTIONS, CORPUS-WIDE OVER 229 DOCS).

Config	H@4	H@10	MRR@10	P-hit
BM25-only	12.7%	25.3%	0.102	0.0%
Vector-only	**26.7%**	**33.3%**	0.175	**24.7%**
Full (default)	22.0%	30.0%	0.164	22.7%
Full (tuned)	25.3%	**33.3%**	**0.179**	22.7%
LightRAG (Gemini)†	–	34.7%‡	–	–
GraphRAG (Gemini)†	–	24.0%‡	–	–

†Reconfigured to OHARA's Gemini models, same 200-doc corpus. ‡Unranked context-blob overlap, not top-k; MRR/P-hit undefined.

TABLE III
MULTIHOP-RAG RETRIEVAL QUALITY (500 STRATIFIED QUERIES).

Config	H@10	MRR@10	P-hit	Null abst.
BM25-only	94.4%	0.727	4.3%*	96.8%*
Vector-only	**98.4%**	0.811	**92.5%**	13.6%
Full (default)	96.5%	0.791	91.5%	**45.6%**
− Corrob.	96.8%	0.795	92.0%†	0.0%
Full (tuned)	**98.4%**	**0.812**	92.0%	31.2%
LightRAG (Gemini)‡	68.0%§	–	–	–
GraphRAG (Gemini)‡	58.4%§	–	–	–

*BM25-only's 4.3% P-hit reflects a near-empty Principal tier (few candidates ever corroborate); its 96.8% abstention is a direct consequence of that emptiness, not discrimination; read both asterisked cells together, not as independent signals. †Proxied as top-5. ‡Same LightRAG+Gemini and GraphRAG+Gemini baselines as Table II, run on the full 608-document MultiHop-RAG corpus (500 queries); GraphRAG uses local search, community level 2. §Not H@10: both baselines' unranked context blobs have no document ids, so we score answer-string presence (does the gold short answer appear in the context) instead of doc-level ranking, a looser, non-comparable metric to this table's other columns.

A. QASPER Retrieval Quality

Table II outlines retrieval configurations. Lexical-only BM25 underperforms vector-only retrieval (25.3% vs. 33.3% Hits@10) due to cross-document term collisions across 229 documents. The default fusion pipeline (30.0% Hits@10) trails vector-only because uncalibrated lexical noise overrides semantic signals. Tuning fusion parameters via a 6-point grid search over a 75-query subset selects (0.6 BM25, 1.0 Vector), then re-run on the full 150 queries, the same 150 questions reported here, with no held-out split. This parity figure therefore carries some tuned-on-test risk. Recovering parity on Hits@10 (33.3% = 33.3%) and reaching 0.179 MRR@10, the unchanged transfer of these weights to MultiHop-RAG (Sec. VII-A, Finding 1) is mitigating evidence against overfitting rather than independent proof.

Decomposing failures against an within-document oracle demonstrates that ∼42% of corpus-wide errors occur during initial macro document location rather than micro paragraph selection. Once the target document is correctly located, paragraph selection succeeds for roughly two-thirds of queries (oracle Hits@10 64.4%).

To address the lack of a head-to-head baseline against cited graph-RAG systems, we ran LightRAG [8] and GraphRAG [7], both reconfigured to use OHARA's exact Gemini models, on the identical corpus. Scored via the same gold-evidence-snippet overlap, LightRAG reaches a 34.7% hit rate and GraphRAG (local search, community level 2) reaches 24.0% (36/150), both under an unranked, uncapped-context protocol (no top-k cutoff, no rank order). The comparison is therefore directional rather than a controlled ranking test, and likely understates OHARA's context-efficiency advantage rather than overstating it. GraphRAG's community-report-mediated retrieval trailing LightRAG's direct entity/chunk retrieval on this corpus is consistent with QASPER's single-hop, within-document question style, which does not exercise GraphRAG's cross-community summarization strength.

B. MultiHop-RAG Retrieval Quality and Abstention

Parameters tuned on QASPER transfer directly to MultiHop-RAG (Table III), reaching Hits@10 and MRR parity while improving MAP (0.556 vs. 0.538). We also extended both the LightRAG+Gemini and GraphRAG+Gemini baselines (Section VII-A) to the full 608-document MultiHop-RAG corpus: LightRAG reaches 68.0% answer-string recall (340/500 queries) and GraphRAG (local search) reaches 58.4% (292/500), both under the same unranked-context caveat as the QASPER baseline: directional evidence, not a controlled comparison to our document-level Hits@10. GraphRAG trailing LightRAG on both corpora, by a wider margin here than on QASPER, is consistent with MultiHop-RAG's cross-document reasoning questions not aligning with local search's single-community retrieval scope; global search was not evaluated due to its substantially higher per-query cost. Empirical evaluation highlights three key insights:

- **Finding 1 (Parameter Transfer):** Fusion weights generalize across distinct domains, confirming that initial performance gaps stemmed from lexical weight bias rather than pipeline architecture.
- **Finding 2 (Corroboration Abstention):** Corroboration gating changes abstention behavior, not ranking quality. The constrained Principal tier abstains on 45.6% of unanswerable queries (57/125, roughly ±8–9pp at typical confidence given n=125) compared to 0.0% for plain top-k cutoffs, maintaining a 91.5% Principal-hit rate on answerable queries. Disabling the constraint moves Hits@10/MRR by under 0.5 points (Table III, row −Corrob.); the entire measured effect is on abstention, not on filtering noisy ranks.
- **Finding 3 (Temporal Decay Boundaries):** Disabling decay scoring slightly improves MRR on temporal query slices (0.749 vs 0.736). Temporal decay acts as a guardrailed recency prior rather than a universal ranking mechanism, as benchmark queries test event ordering rather than publication recency.

End-to-end generation checks (`gemini-2.5-flash-lite`) yield 54% accuracy for tuned fusion versus 57% for vector-only retrieval. Context omission accounts for 41% of failures rather than LLM hallucination.

VIII. Discussion

Limitations. The system relies on Gemini backends for document parsing, tagging, and enrichment. Evaluation is limited to clean English digital text; noisy OCR inputs remain unexamined. While weight parameters transfer successfully between evaluated benchmarks, default settings require broader validation. Interactive visual rendering displays visual clutter beyond $\sim$50 active document discs.

Threats to Validity. Two evaluation-protocol gaps deserve explicit acknowledgment rather than being left implicit in the tuned-on-test and unranked-metric caveats above. First, the QASPER fusion weights (Sec. VII-A) were selected by grid search and re-run on the same 150 queries with no held-out split; unchanged transfer to MultiHop-RAG is mitigating evidence, not a substitute for formal k-fold cross-validation, which we defer to a follow-up study with a larger annotated QASPER subset. Second, the LightRAG baseline (Tables II–III) is scored via unranked answer-string/context recall because LightRAG's output carries no document ids; remapping its retrieved fragments back to source document ids to obtain a directly comparable ranked Hits@10/MRR is left as future work, and until then the LightRAG numbers should be read as directional context-availability evidence, not a controlled ranking comparison.

A further limitation concerns the temporal decay mechanism itself: neither QASPER nor MultiHop-RAG was constructed to isolate recency-sensitive retrieval. MultiHop-RAG's temporal queries probe event-ordering within a fixed, largely static news window, correctly sequencing facts already present in the retrieved context, rather than penalizing outdated documents in favor of fresher ones covering the same entity or topic. Because the corpus lacks paired same-topic documents with meaningfully different publication recency and a ground truth that shifts accordingly, our benchmarks cannot directly exercise the decay parameter's intended behavior, and Finding 3 (§VII-A) should be read as evidence about event-ordering robustness rather than a validation of recency-weighted ranking. This does not undermine decay's role as one of the three coordinate axes (τ, δ) of the unified Space-Time Graph substrate itself (that architectural contribution stands independent of ranking metrics), but it does mean the substrate's recency-prioritization behavior remains architecturally proposed rather than empirically validated. A proper evaluation requires either a constructed slice with same-entity documents of varying age and time-dependent gold answers, or a genuinely live corpus with real freshness constraints (e.g., news feeds or versioned technical documentation, per hypothesis H2). We leave this to future work and caution readers against treating our decay results as more than a guard-rail sanity check.

We are explicit that our central retrieval-quality claim is parity, not superiority: the eight-phase fusion pipeline, five-tuple graph model, and five-layer temporal guard-rail add real architectural complexity for a tuned configuration that ties vector-only retrieval on both corpora (Tables II–III). That complexity is not free, but it does buy three things a single dense-vector index does not: (i) the corroboration-gated abstention behavior of Finding 2, which vector-only and both graph-RAG baselines cannot express at all (0.0% abstention by construction); (ii) ranked, per-signal-attributed provenance rather than an unranked context blob; and (iii) the structural/temporal/ontology substrate the visualization layer depends on. Readers should weigh the paper's contribution accordingly: as an auditability and abstention mechanism layered on top of retrieval that is no worse than the simplest strong baseline, not as a ranking-quality improvement over it.

Research Agenda. We separate empirical findings from open hypotheses:

- **(H1) Dynamic Traversal Re-scoring:** Modify graph traversal parameters to allow node re-scoring, enabling multi-angle corroboration across existing graph paths.
- **(H2) Recency-Sensitive, Event-Aware Decay:** Evaluate decay scoring parameters on live corpora with real freshness constraints (news feeds, active software documentation), paired with an event-extraction layer that decouples narrative event dates from raw publication timestamps, directly targeting the event-chronology confound behind Finding 3.
- **(H3) Human Task Performance:** Execute structured user studies evaluating spatial navigation efficiency against standard search interfaces.
- **(H4) Macro-Micro Rescue for Pre-filtering:** Run a parallel micro-chunk vector pass alongside the coarse document-rollup pre-filter (Sec. III) so high-confidence chunk-level hits can rescue document roots falsely pruned by aggregate tag dilution, directly targeting the $\sim$42% macro-localization failure rate (Sec. VII-A).
- **(H5) Adaptive Visual Level-of-Detail:** Collapse out-of-view document discs into aggregated category spheres to control visual clutter beyond the $\sim$50-document threshold noted above, without increasing WebGL draw calls.

We believe this is just a beginning toward auditable, spatial-temporal knowledge interfaces.

IX. Conclusion

OHARA's Space-Time Graph unifies structural hierarchy, temporal decay, and ontology pivots into a single navigable substrate: a proposed unified coordinate system for data that exists jointly along space and time, of which decay scoring is the temporal-axis instance evaluated here only as a guard-rail sanity check, not a validated recency-ranking mechanism. The framework does not outrank dense vector retrieval or the LightRAG/GraphRAG baselines we evaluated on ranking quality. It ties the former and, under a looser unranked-context metric, sits within range of the latter on QASPER while trailing both on MultiHop-RAG's answer-string recall. What it adds instead is a corroboration-gated Principal tier that abstains on 45.6% of unanswerable queries where every baseline we tested abstains on 0%, plus ranked per-signal provenance neither baseline exposes. Offline ingestion builds the complete semantic graph at $\sim$\$12 per 1,000 documents, cheaper than our GraphRAG baseline runs (\$16–21 per 1,000)

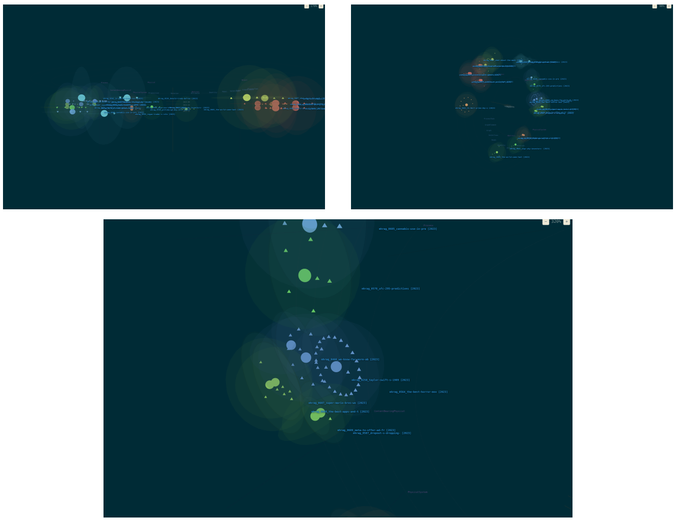

Fig. 2. Three camera views of the same 25-document Space-Time Graph (color = top SUMO category per document). *Top left*: sunburst top-down, looking down $-$Y along the ontology cross-section; documents cluster into SUMO category columns (*Process*, *Physical*, *Object*, *Proposition* labels visible). *Top right*: oblique tunnel view, camera offset along Z so publication time reads as depth. *Bottom*: close-up on a single document disc, showing its section hierarchy as a concentric ring of nodes (triangles) around the document root (sphere).

on the same corpora, and 3D rendering scales sub-linearly. These findings support a narrower claim than raw retrieval superiority: that a structured, auditable knowledge base can match a strong simple baseline on ranking while adding abstention behavior and provenance that flat or purely graph-summarization RAG systems lack.

Acknowledgments

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